

Multiple Choice (10 marks)

1. A DNA molecule has 180 base pairs and is 20 percent adenine. How many cytosine nucleotides are present in this molecule of DNA?
 - (a) 108 nucleotides
 - (b) 120 nucleotides
 - (c) 150 nucleotides
 - (d) 144 nucleotides
 - (e) 60 nucleotides
2. What hormones are involved with the changes that occur in a pubescent female and male?
 - (a) Cortisol, aldosterone, and adrenal androgens
 - (b) Insulin, glucagon, and somatostatin
 - (c) Cholecystokinin, ghrelin, and leptin
 - (d) Vasopressin, erythropoietin, and renin
 - (e) Epinephrine, norepinephrine, and dopamine
3. About 0.5 percent of the American population have epilepsy, the falling sickness. Discusses the genetics of this disease.
 - (a) Epilepsy is the result of a chromosomal aneuploidy, similar to Down syndrome.
 - (b) Epilepsy is always inherited in an X-linked recessive pattern.
 - (c) Epilepsy is a dominant trait.
 - (d) Epilepsy is caused by an autosomal dominant trait.
 - (e) Epilepsy is due to an autosomal recessive trait.
4. In the formation of the earliest cells, which of the following components most likely arose first?
 - (a) Lysosome
 - (b) Chloroplast
 - (c) Ribosome
 - (d) Golgi apparatus
 - (e) Plasma membrane
5. Which of the following characteristics would allow you to distinguish a prokaryotic cell from an animal cell?

- (a) Centrioles
- (b) Chloroplasts
- (c) Lysosomes
- (d) Cell wall
- (e) Mitochondria

True / False (10 marks)

Answer True or False

6. True or False: Polygenic traits are characteristics that are influenced by multiple genes, leading to a range of phenotypes.
7. True or False: Cycads have flowers that can remain dormant for several years before producing seeds, making them unique among seed plants.
8. True or False: Complete equilibrium in a gene pool is not expected due to the influence of factors like mutation and natural selection.
9. True or False: In chick embryos, the yolk sac serves as the extraembryonic membrane that provides nourishment to the developing fetus.
10. True or False: Distantly related organisms should have identical physical characteristics due to their shared ancestry.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the number of chromatids present in an animal with a diploid number of 8 during (a) the tetrad stage of meiosis, (b) late telophase of the first meiotic division, and (c) metaphase of the second meiotic division.
12. Discuss the key environmental factors that influence the characteristics of organisms within an ecosystem, providing examples to support your analysis.

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